

Chapter 4, Exercise 3: QDA Bayes Classifier

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Setup

We consider K classes with a single predictor ($p = 1$). An observation belonging to the k th class is assumed to be drawn from a class-specific normal distribution:

$$X \mid Y = k \sim N(\mu_k, \sigma_k^2)$$

Note that, unlike the LDA setting in Section 4.4.1, we do **not** assume a common variance across classes, each class k has its own variance σ_k^2 .

$$f_k(x) = \frac{1}{\sqrt{2\pi} \sigma_k} \exp \left[-\frac{(x - \mu_k)^2}{2\sigma_k^2} \right]$$

Step 1: Bayes' theorem and the decision rule

By Bayes' theorem, the posterior probability that $X = x$ belongs to class k is:

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)}$$

The Bayes classifier assigns x to the class k that maximizes $p_k(x)$. Since the denominator $\sum_l \pi_l f_l(x)$ does not depend on k , this is equivalent to maximizing the numerator $\pi_k f_k(x)$. Because $\log(\cdot)$ is a strictly increasing function, maximizing $\pi_k f_k(x)$ is equivalent to maximizing $\log(\pi_k f_k(x))$. We therefore define the discriminant function:

$$\delta_k(x) = \log(\pi_k) + \log(f_k(x))$$

Step 2: Expand $\log f_k(x)$

Taking the log of the normal density:

$$\log(f_k(x)) = -\log(\sqrt{2\pi} \sigma_k) - \frac{(x - \mu_k)^2}{2\sigma_k^2}$$

Expanding the square $(x - \mu_k)^2 = x^2 - 2\mu_k x + \mu_k^2$:

$$\log(f_k(x)) = -\frac{1}{2} \log(2\pi) - \log(\sigma_k) - \frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2} x - \frac{\mu_k^2}{2\sigma_k^2}$$

The term $-\frac{1}{2}\log(2\pi)$ is a constant that does not depend on k , so it is the same additive term for every class and can be dropped when comparing classes (it does not affect which class attains the maximum). Substituting the remainder into $\delta_k(x)$:

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2}x - \frac{\mu_k^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k)$$

Step 3: Why this is quadratic, not linear

Write $\delta_k(x)$ in the general polynomial form $\delta_k(x) = a_kx^2 + b_kx + c_k$, where:

$$a_k = -\frac{1}{2\sigma_k^2}, \quad b_k = \frac{\mu_k}{\sigma_k^2}, \quad c_k = -\frac{\mu_k^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k)$$

The key quantity is the coefficient $a_k = -\frac{1}{2\sigma_k^2}$ on the x^2 term. In the LDA setting of Section 4.4.1, all classes share a common variance σ^2 ($\sigma_1 = \dots = \sigma_K = \sigma$), so $a_k = -\frac{1}{2\sigma^2}$ is identical for every class. When comparing two classes k and j by looking at $\delta_k(x) - \delta_j(x)$, the x^2 terms are $a_kx^2 - a_jx^2 = 0$ because $a_k = a_j$ — they cancel exactly, leaving a function that is linear in x . That is precisely why the Bayes/LDA decision boundary in (4.18) is linear.

In this problem, however, we do **not** assume $\sigma_1^2 = \dots = \sigma_K^2$. Each class has its own variance σ_k^2 , so in general $a_k \neq a_j$ for $k \neq j$. Consequently, when we compare two classes via:

$$\delta_k(x) - \delta_j(x) = (a_k - a_j)x^2 + (b_k - b_j)x + (c_k - c_j)$$

the coefficient $(a_k - a_j)$ is generally nonzero, so the x^2 term does **not** cancel. This means:

1. $\delta_k(x)$ itself is a quadratic (not linear) function of x , since it contains a nonzero x^2 coefficient $a_k = -\frac{1}{2\sigma_k^2}$.
2. The decision boundary between any two classes — the set of x where $\delta_k(x) = \delta_j(x)$ — is the solution to a quadratic equation $(a_k - a_j)x^2 + (b_k - b_j)x + (c_k - c_j) = 0$, which can have up to two real roots (i.e., up to two boundary points), unlike the single boundary point produced by a linear equation.
3. This is exactly the QDA setup of Section 4.4.3: allowing class-specific variances (or, in the multivariate case, class-specific covariance matrices Σ_k) introduces the quadratic term $x^\top \Sigma_k^{-1}x$, which is what gives QDA its name and its curved decision boundaries.

Conclusion

Because the Bayes classifier assigns x to the class maximizing

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2}x - \frac{\mu_k^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k),$$

and this expression is a genuine quadratic function of x whenever the class variances σ_k^2 differ, the Bayes classifier is not linear in general; it is quadratic. Linearity (as in Section 4.4.1's LDA derivation) is a special case that only arises under the additional restriction $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_K^2$.